Iniziato	martedì, 8 luglio 2025, 14:43
Stato	Completato
Terminato	martedì, 8 luglio 2025, 14:58
Tempo impiegato	15 min.

Domanda 1

Risposta corretta

Punteggio max.: 2,00

Let f, g, p be the functions such that $f \in O(g)$ and $g \in O(p)$. Then, necessarily:

Scegli una o più alternative:

- [ ] $f \in \Theta(g)$
- [ ] $p \in O(f)$
- [x] $f \in O(p)$ ✔

Your answer is correct.

Domanda 2

Risposta corretta

Punteggio max.: 2,00

Which of the following sets can be easily encoded as binary strings?

Scegli una o più alternative:

- [x] Vectors of rational numbers ✔
- [x] Undirected graphs ✔
- [x] Strings on an alphabet of size 10^2 ✔
- [ ] Vectors of complex numbers

Your answer is correct.

Domanda 3

Risposta corretta

Punteggio max.: 2,00

The problem of deciding whether an undirected graph $\mathbb{G}$ contains an independent set of size equal to the size of $\mathbb{G}$ is:

Scegli una o più alternative:

- [x] In the class $\mathbf{EXP}$ ✔
- [ ] Not in the class $\mathbf{NP}$
- [ ] Undecidable, and $\mathbf{NP}$-hard.
- [x] Decidable in polynomial time ✔

Your answer is correct.

Domanda 4

Parzialmente corretta

Punteggio max.: 2,00

Which ones of the following inclusions between complexity classes is compatible with our current knowledge (i.e. maybe we do not know if this is true, but it could be)? Here, $\mathbf{NPC}$ stands for the class of all $\mathbf{NP}$-complete problems, while $\mathbf{NPNC}$ stands for the class of all $\mathbf{NP}$ problems which are not complete.

Scegli una o più alternative:

- [ ] $\mathbf{NP} \subseteq \mathbf{NPNC}$.
- [ ] $\mathbf{NPC} \subseteq \mathbf{P}$.
- [x] $\mathbf{NP} \subseteq \mathbf{NPC}$. ✔
- [x] $\mathbf{EXP} \subseteq \mathbf{NPC}$. ✔

Your answer is partially correct.

Hai selezionato correttamente 2.

Domanda 5

Risposta corretta

Punteggio max.: 2,00

The concept classes of axis-aligned squares in $\mathbb{R}^2$ is:

Scegli una o più alternative:

- [x] PAC learnable ✔
- [ ] Not PAC learnable
- [x] Efficiently PAC learnable, having finite VC-dimension. ✔
- [ ] PAC learnable, but not efficiently so, having infinite VC-dimension.

Your answer is correct.

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